

DANMARKS TEKNISKE UNIVERSITET



MODELING CO₂ CONCENTRATIONS USING HIDDEN
MARKOV MODELS

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SPECIAL COURSE IN STATISTICAL MODELLING

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Model	#Params	LogLik	AIC	BIC
HMM(1)	20	-10388.87	20815.73	20918.68
AR(1), HMM(1)	24	-9359.73	18765.45	18890.06
AR(2), HMM(1)	28	-9273.19	18600.38	18746.64
AR(1), HMM(2)	84	-9130.90	18427.80	18877.46
AR(2), HMM(2)	100	-9045.40	18288.79	18825.07
Covariates-HMM(1)	44	-10229.71	20545.41	20778.39
AR(1), Covariates-HMM(1)	48	-9272.25	18638.51	18893.13
AR(2), Covariates-HMM(1)	52	-9192.95	18487.90	18764.16

Tabel 1: Per-model log-likelihood, AIC and BIC for the fitted HMMs.

Base Model	Expanded Model	LRT	df	p-value	Δ AIC	Δ BIC
HMM(1)	AR(1), HMM(1)	2058.28	5	0.0000	-2050.28	-2028.61
AR(1), HMM(1)	AR(2), HMM(1)	173.07	5	0.0000	-165.07	-143.40
AR(1), HMM(1)	AR(1), HMM(2)	457.65	61	0.0000	-337.65	-12.60
AR(2), HMM(1)	AR(2), HMM(2)	455.59	73	0.0000	-311.59	78.43
AR(1), HMM(2)	AR(2), HMM(2)	171.01	17	0.0000	-139.01	-52.34
HMM(1)	Covariates-HMM(1)	318.32	25	0.0000	-270.32	-140.29
Covariates-HMM(1)	AR(1), Covariates-HMM(1)	1914.90	5	0.0000	-1906.90	-1885.23
AR(1), Covariates-HMM(1)	AR(2), Covariates-HMM(1)	158.61	5	0.0000	-150.61	-128.94
AR(1), HMM(1)	AR(1), Covariates-HMM(1)	174.95	25	0.0000	-126.95	3.08
AR(2), HMM(1)	AR(2), Covariates-HMM(1)	160.49	25	0.0000	-112.49	17.52

Tabel 2: Likelihood-ratio tests for each nested pair from the model hierarchy (Figure ??). Negative Δ AIC/ Δ BIC favour the expanded model.